

PILOT MOUND, MB, Canada

WMO: 711480

Lat: 49.1881N

Lon: 98.9064W

Elev: 475

StdP: 95.75

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-28.9	-35.5	0.1	-30.9	-32.7	0.2	-28.5	13.3	-12.4	12.3	-12.1	3.3	290	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.2	21.0	27.5	20.0	25.9	18.8	23.2	27.0	21.7	25.8	20.4	24.3	5.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.6	25.8	20.3	15.8	24.2	18.9	14.5	22.8	71.5	27.2	65.8	25.9	60.9	24.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.5	9.4	DB	-34.5	33.1	2.5	1.7	-36.3	34.3	-37.7	35.2	-39.1	36.2	-40.9	37.4	(4)
(5)			WB	-34.8	25.3	2.8	1.6	-36.8	26.5	-38.4	27.4	-40.0	28.3	-42.0	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-15.4	-13.6	-6.2	3.2	10.4	15.9	18.3	17.6	12.8	4.8	-4.5	-12.4	(6)	
	DBStd	13.48	8.12	7.67	7.68	6.01	4.77	3.57	3.00	3.33	4.58	5.12	6.74	7.64	(7)	
	HDD10.0	3549	788	661	503	213	55	3	0	0	22	176	434	694	(8)	
	HDD18.3	5815	1046	894	761	454	249	88	38	54	173	420	684	953	(9)	
	CDD10.0	869	0	0	1	9	66	180	257	235	107	14	0	0	(10)	
	CDD18.3	94	0	0	0	0	3	15	37	31	8	1	0	0	(11)	
	CDH23.3	1088	0	0	0	2	60	165	351	376	126	9	0	0	(12)	
	CDH26.7	238	0	0	0	0	12	26	65	104	29	2	0	0	(13)	
(14) Wind	WSAvg	4.5	4.6	4.7	4.9	5.1	5.2	4.3	3.6	3.8	4.4	4.8	4.7	4.5	(14)	
(15) Precipitation	PrecAvg	495	15	13	18	26	74	91	75	68	42	34	19	19	(15)	
	PrecMax	709	30	40	61	55	159	193	166	176	118	142	62	47	(16)	
	PrecMin	320	0	0	1	1	32	30	18	17	2	2	0	1	(17)	
	PrecStd	98	8	10	14	14	33	36	32	37	23	28	17	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	4.6	14.5	22.5	28.2	29.9	30.5	32.4	30.2	24.3	14.2	5.1	(19)	
		MCWB	1.8	2.5	10.1	12.3	16.5	21.5	24.7	21.0	19.4	14.7	8.8	2.4	(20)	
	2%	DB	1.0	1.6	8.8	19.0	24.8	27.0	28.4	29.4	26.8	19.2	9.9	1.9	(21)	
		MCWB	-0.1	0.4	5.7	10.7	13.6	19.3	22.5	20.7	17.6	12.1	5.8	0.6	(22)	
	5%	DB	-1.2	0.1	5.2	16.2	22.1	25.3	26.9	27.3	24.3	15.9	6.7	-0.4	(23)	
		MCWB	-1.8	-0.6	3.3	9.4	12.9	18.1	21.3	19.7	16.5	10.6	4.1	-1.2	(24)	
	10%	DB	-3.7	-2.2	2.6	13.1	19.5	23.6	25.3	25.2	21.4	12.9	3.9	-2.3	(25)	
		MCWB	-4.3	-3.0	1.4	7.9	12.1	17.2	20.1	18.5	15.0	9.2	2.3	-2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	3.0	10.3	14.2	18.2	24.3	26.2	24.2	21.3	15.7	9.5	2.7	(27)	
		MCDB	3.3	4.5	14.2	19.1	24.2	27.0	29.2	28.5	27.2	22.1	14.2	4.7	(28)	
	2%	WB	0.1	0.7	6.2	11.7	16.2	21.6	23.8	22.2	19.1	13.4	6.3	0.8	(29)	
		MCDB	1.0	1.5	9.0	17.5	20.8	25.2	27.2	27.4	24.1	18.4	8.9	2.1	(30)	
	5%	WB	-1.8	-0.6	3.5	9.9	14.7	19.6	22.3	20.8	17.5	11.4	4.1	-1.0	(31)	
		MCDB	-1.1	0.3	5.4	15.5	19.4	23.3	26.0	25.6	22.2	15.1	6.3	-0.2	(32)	
	10%	WB	-4.0	-2.8	1.7	8.2	13.2	18.0	21.0	19.5	16.0	9.4	2.3	-2.7	(33)	
		MCDB	-3.5	-2.1	2.8	13.0	18.0	22.1	24.5	23.8	20.9	12.8	3.8	-2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	9.2	11.1	12.4	11.0	11.6	12.9	12.7	10.4	8.8	8.6	(35)	
		MCDBR	11.0	10.2	11.2	16.8	17.2	13.9	13.7	15.9	17.0	16.2	13.2	10.5	(36)	
	5% WB	MCWBR	10.1	9.1	8.3	9.5	8.7	7.6	8.5	8.3	8.8	9.6	9.5	9.1	(37)	
		MCWBR	10.8	10.2	11.0	15.4	12.9	11.7	12.7	13.7	14.2	14.5	12.2	10.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.267	0.276	0.290	0.352	0.366	0.372	0.379	0.378	0.345	0.293	0.264	0.244	(40)	
		taud	2.275	2.273	2.411	2.326	2.343	2.366	2.353	2.346	2.424	2.548	2.486	2.402	(41)	
	Ebn at Noon		790	876	930	894	893	887	875	858	856	860	810	789	(42)	
		Edh at Noon	71	91	95	116	119	117	117	113	94	70	58	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.52	2.69	3.93	4.99	5.58	6.01	6.37	5.24	3.83	2.36	1.35	1.08	(44)	
	RadStd		0.12	0.22	0.32	0.36	0.41	0.40	0.28	0.35	0.28	0.30	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.62	N/A	N/A	N/A	N/A

Nomenclature: See separate page